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Batch: E21  
EDS ASSIGNMENT NO.2

20 problem statements for a given dataset using NumPy and Pandas and Apply NumPy and Pandas methods to find the solution for the formulated problem statements.

DATASET: Sales Dataset

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Sales Dataset Problem Statements

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RAM Disk

[4] 8s

from google.colab import files  
uploaded = files.upload()

Chocolate Sales (1).csv  
Chocolate Sales (1).csv(text/csv) - 212462 bytes, last modified: n/a - 100% done  
Saving Chocolate Sales (1).csv to Chocolate Sales (1).csv

[5] 0s

import pandas as pd  
import numpy as np  
  
df = pd.read\_csv("Chocolate Sales (1).csv")

[ ]

# 1> Problem Statement: Find which salesperson generated the highest total sales amount.  
top\_salesperson = df.groupby('Sales Person')['Amount'].sum().sort\_values(ascending=False)  
print(top\_salesperson.head(1))

Sales Person  
Dotty Strutley    \$91.00\$280.00\$16,114.00\$273.00\$9,310.00\$3,108....  
Name: Amount, dtype: object

[ ]

# 2> Problem Statement: Identify the salesperson with the lowest sales.  
lowest\_salesperson = df.groupby('Sales Person')['Amount'].sum().sort\_values()  
print(lowest\_salesperson.head(1))

... Sales Person  
Curtice Advani    \$1,085.00\$6,979.00\$5,243.00\$4,865.00\$854.00\$85...  
Name: Amount, dtype: object

[ ]

# 3> Problem Statement: Determine which country contributes the most revenue.  
country\_sales = df.groupby('Country')['Amount'].sum().sort\_values(ascending=False)  
print(country\_sales.head(1))

Country  
India    \$7,896.00\$4,501.00\$5,376.00\$168.00\$2,443.00\$6,...  
Name: Amount, dtype: object

[ ]

# 4> Problem Statement: Find the most sold product.  
product\_sales = df.groupby('Product')['Amount'].sum().sort\_values(ascending=False)  
print(product\_sales.head(1))

Variables

Terminal

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Python 3

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Sales Dataset Problem Statements

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Product  
White Choc \$9,492.00\$2,800.00\$9,681.00\$7,126.00\$1,043.00\$...  
Name: Amount, dtype: object

```
[ ]
# 5> Problem Statement: Determine which product sells the most in every country.
best_product_country = df.groupby(['Country', 'Product'])['Amount'].sum().reset_index()
best_product_country = best_product_country.loc[
    best_product_country.groupby('Country')['Amount'].idxmax()
]
print(best_product_country)
```

|     | Country     | Product \            |
|-----|-------------|----------------------|
| 9   | Australia   | Drinking Coco        |
| 43  | Canada      | White Choc           |
| 59  | India       | Orange Choco         |
| 87  | New Zealand | White Choc           |
| 96  | UK          | Choco Coated Almonds |
| 126 | USA         | Organic Choco Syrup  |

  

|     | Amount                                                   |
|-----|----------------------------------------------------------|
| 9   | \$91.00\$1,603.00\$623.00\$11,550.00\$2,807.00\$6,83...  |
| 43  | \$9,275.00\$420.00\$3,381.00\$6,587.00\$4,067.00\$4,...  |
| 59  | \$9,408.00\$4,053.00\$3,122.00\$784.00\$1,190.00\$4,...  |
| 87  | \$9,492.00\$2,800.00\$7,126.00\$3,171.00\$4,361.00\$...  |
| 96  | \$9,324.00\$4,333.00\$4,270.00\$10,164.00\$3,010.00\$... |
| 126 | \$9,870.00\$16,233.00\$7,721.00\$2,380.00\$2,331.00\$... |

```
[ ]
# 6> Problem Statement: Find the date on which maximum sales occurred.
peak_day = df.groupby('Date')['Amount'].sum().sort_values(ascending=False)
print(peak_day.head(1))
```

```
... Date
20/06/2023 $970.96$1,835.42$3,133.42$8,895.70$1,264.43$16...
```

```
[ ]
# 7> Problem Statement: Find which salesperson performs best in which country.
matrix = pd.pivot_table(
    df,
    values='Amount',
    index='Sales Person',
    columns='Country',
    aggfunc='sum',
    fill_value=0
)
```

Variables

Terminal

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Python 3

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Sales Dataset Problem Statements

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[ ]

print(matrix)

Country Sales Person Australia \

|                     |             |             |             |             |             |            |
|---------------------|-------------|-------------|-------------|-------------|-------------|------------|
| Andria Kimpton      | \$4,438.00  | \$1,603.00  | \$4,186.00  | \$6,678.00  | \$9,744.00  | \$...      |
| Barr Faughny        | \$15,421.00 | \$8,113.00  | \$2,933.00  | \$8,498.00  | \$2,303.00  | \$...      |
| Beverie Moffet      | \$6,790.00  | \$1,827.00  | \$735.00    | \$3,192.00  | \$2,240.00  | \$10...    |
| Brien Boise         | \$2,835.00  | \$6,888.00  | \$8,575.00  | \$12,516.00 | \$5,124.00  | \$...      |
| Camilla Castle      | \$3,654.00  | \$6,048.00  | \$3,178.00  | \$4,669.00  | \$637.00    | \$7,...    |
| Ches Bonnell        | \$4,116.00  | \$1,043.00  | \$9,660.00  | \$3,136.00  | \$7,490.00  | \$...      |
| Curtice Advani      | \$6,979.00  | \$854.00    | \$3,472.00  | \$2,912.00  | \$4,676.00  | \$50...    |
| Dennison Crosswaite | \$5,859.00  | \$6,881.00  | \$5,740.00  | \$2,758.00  | \$7,287.00  | \$...      |
| Dotty Strutley      | \$91.00     | \$280.00    | \$3,108.00  | \$6,979.00  | \$6,790.00  | \$8,225... |
| Gigi Bohling        | \$2,807.00  | \$5,523.00  | \$8,001.00  | \$6,524.00  | \$910.00    | \$7,...    |
| Gunar Cockshoot     | \$3,080.00  | \$6,013.00  | \$3,752.00  | \$602.00    | \$2,317.00  | \$12...    |
| Husein Augar        | \$8,897.00  | \$7,161.00  | \$14,889.00 | \$6,993.00  | \$7,959.00  | \$...      |
| Jan Morforth        | \$12,726.00 | \$3,472.00  | \$3,199.00  | \$3,402.00  | \$5,222.00  | \$...      |
| Jehu Rudeforth      | \$5,194.00  | \$13,706.00 | \$5,691.00  | \$2,611.00  | \$10,486.0  | \$...      |
| Kaine Padly         | \$7,672.00  | \$2,765.00  | \$5,460.00  | \$6,706.00  | \$7,434.00  | \$...      |
| Karlen McCaffrey    | \$7,357.00  | \$7,294.00  | \$238.00    | \$2,303.00  | \$6,678.00  | \$99...    |
| Kelci Walkden       | \$7,273.00  | \$6,916.00  | \$3,185.00  | \$11,116.00 | \$8,715.00  | \$...      |
| Madelene Upcott     | \$7,182.00  | \$5,579.00  | \$7,350.00  | \$2,058.00  | \$1,575.00  | \$...      |
| Mallorie Waber      | \$2,779.00  | \$14,658.00 | \$2,541.00  | \$7,882.00  | \$5,439.00  | \$...      |
| Marney O'Brien      | \$273.00    | \$2,464.00  | \$3,745.00  | \$13,125.00 | \$11,298.00 | \$...      |
| Oby Sorrel          | \$6,034.00  | \$1,043.00  | \$7,672.00  | \$6,944.00  | \$721.00    | \$3,...    |
| Rafaelita Blaksland | \$11,550.00 | \$3,010.00  | \$994.00    | \$2,786.00  | \$4,396.00  | \$3...     |
| Roddy Speechley     | \$6,797.00  | \$4,466.00  | \$12,565.00 | \$4,200.00  | \$3,647.00  | \$...      |
| Van Tuxwell         | \$19,453.00 | \$7,910.00  | \$5,936.00  | \$112.00    | \$644.00    | \$13,...   |
| Wilone O'Kielt      | \$4,284.00  | \$2,030.00  | \$1,743.00  | \$623.00    | \$3,325.00  | \$39...    |

Country Sales Person Canada \

|                     |             |             |             |             |             |         |
|---------------------|-------------|-------------|-------------|-------------|-------------|---------|
| Andria Kimpton      | \$2,163.00  | \$7,154.00  | \$8,204.00  | \$308.00    | \$4,858.00  | \$1,... |
| Barr Faughny        | \$2,100.00  | \$10,185.00 | \$3,402.00  | \$1,141.00  | \$10,969.0  | \$...   |
| Beverie Moffet      | \$8,799.00  | \$9,275.00  | \$13,447.00 | \$14,287.00 | \$8,722.0   | \$...   |
| Brien Boise         | \$16,793.00 | \$8,106.00  | \$4,781.00  | \$10,241.00 | \$5,033.0   | \$...   |
| Camilla Castle      | \$9,716.00  | \$7,770.00  | \$12,481.00 | \$10,095.71 | \$8,671.2   | \$...   |
| Ches Bonnell        | \$1,876.00  | \$7,462.00  | \$5,327.00  | \$15,547.00 | \$9,870.00  | \$...   |
| Curtice Advani      | \$4,865.00  | \$854.00    | \$4,193.00  | \$2,660.00  | \$12,586.00 | \$5...  |
| Dennison Crosswaite | \$2,527.00  | \$455.00    | \$10,479.00 | \$10,885.00 | \$1,750.00  | \$...   |
| Dotty Strutley      | \$4,347.00  | \$11,550.00 | \$252.00    | \$4,849.30  | \$12,048.56 | \$...   |
| Gigi Bohling        | \$1,505.00  | \$9,989.00  | \$11,956.00 | \$3,052.00  | \$3,549.00  | \$...   |
| Gunar Cockshoot     | \$2,058.00  | \$13,888.00 | \$7,756.00  | \$2,744.00  | \$3,024.00  | \$...   |
| Husein Augar        | \$3,696.00  | \$9,772.00  | \$5,131.00  | \$3,381.00  | \$4,049.61  | \$...   |
| Jan Morforth        | \$1,064.00  | \$9,541.00  | \$8,904.00  | \$4,515.00  | \$5,894.00  | \$...   |
| Jehu Rudeforth      | \$441.00    | \$1,848.00  | \$5,460.00  | \$7,203.00  | \$3,857.00  | \$4,... |
| Kaine Padlv         | \$3.269.00  | \$8.687.00  | \$1.288.00  | \$10.213.00 | \$1.729.00  | \$...   |

Variables Terminal

10:06 AM Python 3

|   |   |     |                     |            |             |             |            |             |         |
|---|---|-----|---------------------|------------|-------------|-------------|------------|-------------|---------|
| ⌂ | ⌵ | ... | Karlen McCaffrey    | \$2,919.00 | \$3,017.00  | \$3,479.00  | \$2,415.00 | \$6,328.00  | ...     |
|   |   |     | Kelci Walkden       | \$3,073.00 | \$4,830.00  | \$16,016.00 | \$7,532.00 | \$8,099.00  | ...     |
|   |   |     | Madelene Upcott     | \$7,602.00 | \$10,927.00 | \$7,504.00  | \$3,458.00 | \$14,763.00 | ...     |
|   |   |     | Mallorie Waber      | \$1,442.00 | \$4,781.00  | \$1,428.00  | \$3,577.00 | \$1,435.00  | ...     |
|   |   |     | Marney O'Brien      | \$2,639.00 | \$2,436.00  | \$3,528.00  | \$2,723.00 | \$1,372.00  | ...     |
|   |   |     | Oby Sorrel          | \$3,605.00 | \$1,232.00  | \$1,750.00  | \$6,615.00 | \$12,425.00 | ...     |
|   |   |     | Rafaelita Blaksland | \$7,119.00 | \$9,884.00  | \$483.00    | \$5,096.00 | \$2,520.00  | \$8,... |
|   |   |     | Roddy Speechley     | \$1,652.00 | \$609.00    | \$4,032.00  | \$8,148.00 | \$9,338.00  | \$6,... |
|   |   |     | Van Tuxwell         | \$4,067.00 | \$3,213.00  | \$420.00    | \$8,617.00 | \$1,267.00  | \$6,... |
|   |   |     | Wilone O'Kieft      | \$4,991.00 | \$3,437.00  | \$5,263.45  | \$3,589.18 | \$5,594.84  | ...     |

[illegible]

|         |               |
|---------|---------------|
| Country | New Zealand \ |
|---------|---------------|

[illegible]

```
print(matrix)
```

[illegible]





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Sales Dataset Problem Statements

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Sales Person  
Andria Kimpton 117  
Barr Faughny 129  
Beverie Moffet 150  
Brien Boise 159  
Camilla Castle 96  
Ches Bonnell 144  
Curtice Advani 138  
Dennison Crosswaite 147  
Dotty Strutley 108  
Gigi Bohling 141  
Gunar Cockshoot 129  
Husein Augar 114  
Jan Morforth 117  
Jehu Rudeforth 129  
Kaine Padly 135  
Karlen McCaffrey 141  
Kelci Walkden 162  
Madelene Upcott 135  
Mallorie Waber 123  
Marney O'Brien 135  
Oby Sorrel 147  
Rafaelita Blaksland 102  
Roddy Speechley 129  
Van Tuxwell 153  
Wilone O'Kielt 102  
dtype: int64

# 10> Problem Statement: Count orders for each country.  
orders\_per\_country = df.groupby('Country').size()  
print(orders\_per\_country)

Country  
Australia 615  
Canada 525  
India 552  
New Zealand 519  
UK 534  
USA 537  
dtype: int64

# 11> Problem Statement: Count orders per product.  
orders\_per\_product = df.groupby('Product').size()  
print(orders\_per\_product)

Variables Terminal

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|                      |     |
|----------------------|-----|
| Product              |     |
| 50% Dark Bites       | 180 |
| 70% Dark Bites       | 126 |
| 85% Dark Bars        | 150 |
| 99% Dark & Pure      | 147 |
| After Nines          | 150 |
| Almond Choco         | 144 |
| Baker's Choco Chips  | 123 |
| Caramel Stuffed Bars | 129 |
| Choco Coated Almonds | 117 |
| Drinking Coco        | 168 |
| Eclairs              | 180 |
| Fruit & Nut Bars     | 150 |
| Manuka Honey Choco   | 135 |
| Milk Bars            | 147 |
| Mint Chip Choco      | 135 |
| Orange Choco         | 141 |
| Organic Choco Syrup  | 156 |
| Peanut Butter Cubes  | 147 |
| Raspberry Choco      | 144 |
| Smooth Sliky Salty   | 177 |
| Spicy Special Slims  | 162 |
| White Choc           | 174 |
| dtype: int64         |     |

```
# 12> Problem Statement: Calculate average sales amount per salesperson.
avg_salesperson = df.groupby('Sales Person')['Amount'].mean()
print(avg_salesperson)
```

```
[10] # 13> Problem Statement: Find maximum sale per salesperson.
print(df.groupby('Sales Person')['Amount'].max())
```

|                     |            |
|---------------------|------------|
| Sales Person        |            |
| Andria Kimpton      | \$970.96   |
| Barr Faughny        | \$9,876.86 |
| Beverie Moffet      | \$994.00   |
| Brien Boise         | \$9,771.93 |
| Camilla Castle      | \$9,986.05 |
| Ches Bonnell        | \$9,870.00 |
| Curtice Advani      | \$933.45   |
| Dennison Crosswaite | \$952.00   |
| Dotty Strutley      | \$94.19    |
| Gigi Bohling        | \$949.18   |
| Gunar Cockshoot     | \$9,759.53 |
| Husein Augar        | \$960.69   |

```
Curlice Advani          $935.43
Dennison Crosswaite      $952.00
Dotty Strutley           $94.19
Gigi Bohling             $949.18
Gunar Cockshoot          $9,759.53
Husein Augar             $960.69
Jan Morforth             $99.50
Jehu Rudeforth           $9,851.62
Kaine Padly              $9,928.10
Karlen McCaffrey         $994.00
Kelci Walkden            $975.67
Madelene Upcott          $9,672.76
Mallorie Waber           $969.15
Marney O'Brien           $973.00
Oby Sorrel               $9,928.42
Rafaelita Blaksland      $994.00
Roddy Speechley           $9,976.22
Van Tuxwell              $987.00
Wilone O'Kieft           $9,950.14
Name: Amount, dtype: object
```

```
[13]: # 14> Problem Statement: Find minimum sale per salesperson.
print(df.groupby('Sales Person')['Amount'].min())

>
Product
White Choc    $9,492.00$2,800.00$9,681.00$7,126.00$1,043.00$...
Name: Amount, dtype: object
```

```
[13]: # 15> Problem Statement: Find country generating least revenue.
print(df.groupby('Country')['Amount'].sum().sort_values().head(1))

... Country
Canada    $1,442.00$4,067.00$8,799.00$1,652.00$2,163.00$...
Name: Amount, dtype: object
```

```
[14]: # 16> Problem Statement: Analyze total sales per day.
print(df.groupby('Date')['Amount'].sum())

>
Date
01/02/2022    $1,540.00$3,374.00$10,171.00$9,989.00$2,464.00...
01/02/2023    $1,775.18$3,782.83$10,338.27$10,265.97$2,525.9...
01/02/2024    $1,634.54$4,005.08$11,815.59$11,705.99$2,797.1...
01/03/2022                                $5,229.00$15,008.00
01/03/2023                                $5,747.66$15,653.44
01/05/2023    $5,064.00$664.00$2,440.00$10,573.50$11,102.47$...
```

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Sales Dataset Problem Statements

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31/05/2023 \$5,064.88\$664.82\$2,418.93\$10,573.50\$11,103.47\$...

31/05/2024 \$5,055.05\$631.02\$2,662.54\$11,264.45\$11,863.69\$...

31/08/2022 \$5,614.00

31/08/2023 \$5,674.70

31/08/2024 \$6,170.80

Name: Amount, Length: 504, dtype: object

[21]

# 17> Problem Statement: Identify most active salesperson.  
print(df['Sales Person'].value\_counts().head(1))

Sales Person  
Kelci Walkden 162  
Name: count, dtype: int64

[31]

# 18> Problem Statement: Find country with most transactions.  
print(df['Country'].value\_counts().head(1))

Country  
Australia 615  
Name: count, dtype: int64

[35]

# 19> Problem Statement: Which salesperson sold the highest number of different products?  
print(df.groupby('Sales Person')['Product'].nunique().sort\_values(ascending=False).head(1))

Sales Person  
Kelci Walkden 22  
Name: Product, dtype: int64

[36]

# 20> Problem Statement: Which country has the highest average order value?  
print(df.groupby('Country')['Amount'].mean().sort\_values(ascending=False).head(1))

Country  
Australia NaN  
Name: Amount, dtype: float64

Variables Terminal

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